PART I

Pitch

Every note is higher or lower than another. This part starts with the keyboard you can see and touch, moves to the staff where those keys are written, and ends with the families of notes, scales and keys, that whole pieces are built from.

1 **Keyboard** high and low, the patterns › 2 **Staff** lines, spaces and clefs › 3 **Note names** reading every note › 4 **Sharps and flats** the black keys › 5 **Intervals** the distance between notes › 6 **Scales and keys** notes in families

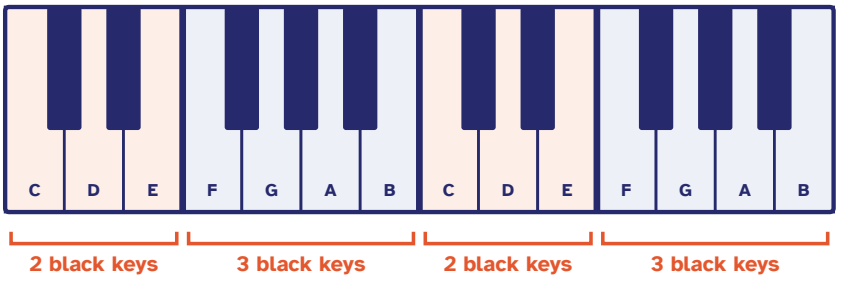
IN THIS PART

The keyboard 22
The staff and the clefs 23
Reading the notes 24
The staff and the keyboard 25
Sharps, flats and naturals 26
Counting intervals 27
Major, minor and perfect 28
Major scales 29
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Keys and key signatures 31
Finding the key 32
The circle of fifths 33

PART I · PITCH · 1

The keyboard

Keys to the **left** sound **lower**; keys to the **right** sound **higher**. The black keys come in groups of **two** and **three**, and that pattern repeats all the way along: it is how you find every note.



The white keys

Seven letters, **A B C D E F G**, then the letters start again, like the days of the week. **C D E** sit around each group of two black keys (C just left of it), **F G A B** around each group of three (F just left of it): the two tints on the picture.

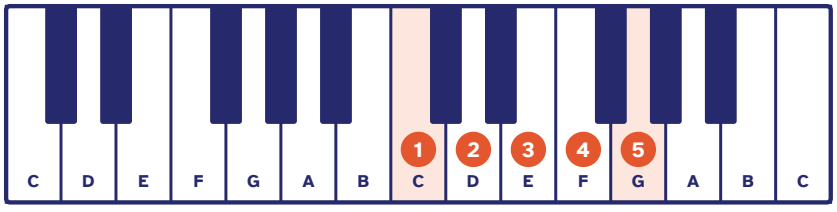
Middle C and octaves

Middle C is the C nearest the middle of the keyboard. From one C to the next is an **octave**: eight white keys, counting both. Octaves are numbered from the bottom: middle C is **C4**, the C below it C3, the C above C5; every key up to the next B shares the number (D4, E4 … B4).

TERM	MEANING	FIRST MET
pitch	how high or low a sound is	Level 0
register	a part of the keyboard: low, middle or high	Level 0
semitone	from one key to the very next key, black or white	Level 0
tone	two semitones	Level 0
octave	from one note to the next with the same name (8 white keys, 12 semitones)	Level 0

IN ACTION

Find G4 on the keyboard



1. **Find middle C (C4):** the C just left of the two black keys nearest the middle.
2. **Count up the letters**, one white key each: C D E F G is five keys.
3. **Same octave**, so the number stays 4.

The key: **G4**. One octave higher, the next G, is G5.

SEE ALSO
The staff p. 23 · Staff and keyboard p. 25

PRACTISE IT
Workbook Pitch 1 · Online: Keyboard patterns

PART I · PITCH · 2

The staff and the clefs

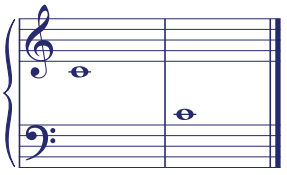
Music is written on a **staff** (or stave) of five lines and four spaces. Lines and spaces are counted from the **bottom**. Higher notes sit higher on the staff. A **clef** at the start tells you which notes the lines and spaces stand for.

CLEF	NAME	HOW IT SHOWS ITS NOTE	FIRST MET
𝄞	**treble clef** (G clef)	curls round the line for G (line 2)	Level 0
𝄢	**bass clef** (F clef)	its two dots sit either side of the line for F (line 4)	Level 0

Other clefs. The alto clef (viola) and the tenor clef (cello and bassoon up high) are C clefs: the line through their centre is middle C. Pianists meet them in theory exams and in chamber music scores (Level 2).

The grand staff

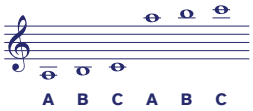
Piano music uses two staves joined by a brace: the treble staff, usually for the right hand, above the bass staff, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.



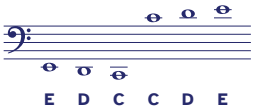
Middle C in the treble staff (left) and in the bass staff (right): the same key.

Ledger lines

Notes above or below the staff sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Count on from the staff as if the lines went on.



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

Octave signs

SIGN	MEANING	FIRST MET
8va	play an octave higher than written, as far as the dotted line goes	Level ½
8vb	play an octave lower	Level ½
15ma	play two octaves higher	Level 4
loco	back where it is written	Level ½

SEE ALSO

Reading the notes p. 24 · The staff and the keyboard p. 25

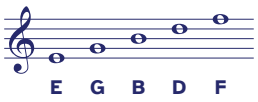
PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

PART I · PITCH · 3

Reading the notes

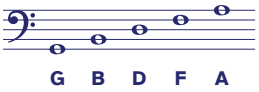
Each line and space has one letter name. Learn them in two ways: by the patterns below, and by a few **landmark notes** you can find instantly and count from.



Treble clef: the lines



Treble clef: the spaces



Bass clef: the lines



Bass clef: the spaces

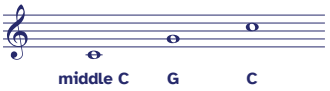
STAFF	NOTES, BOTTOM TO TOP	TO REMEMBER
Treble lines	E G B D F	*Every Good Boy Deserves Fun*
Treble spaces	F A C E	spells *FACE*
Bass lines	G B D F A	*Great Big Dragon Flies Away*
Bass spaces	A C E G	*All Cows Eat Grass*

Landmark notes

Find these first. Every other note is a **step** (line to space) or a **skip** (line to line) away.
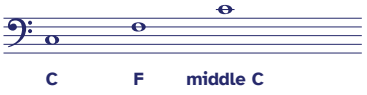
Treble clef

Middle C on its ledger line below; the G the clef curls round (line 2); the C in the third space.



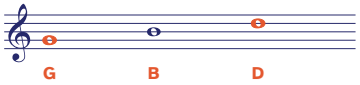
Bass clef

The C in the second space; the F between the clef's dots (line 4); middle C on its ledger line above.



IN ACTION

Read a note you are not sure of



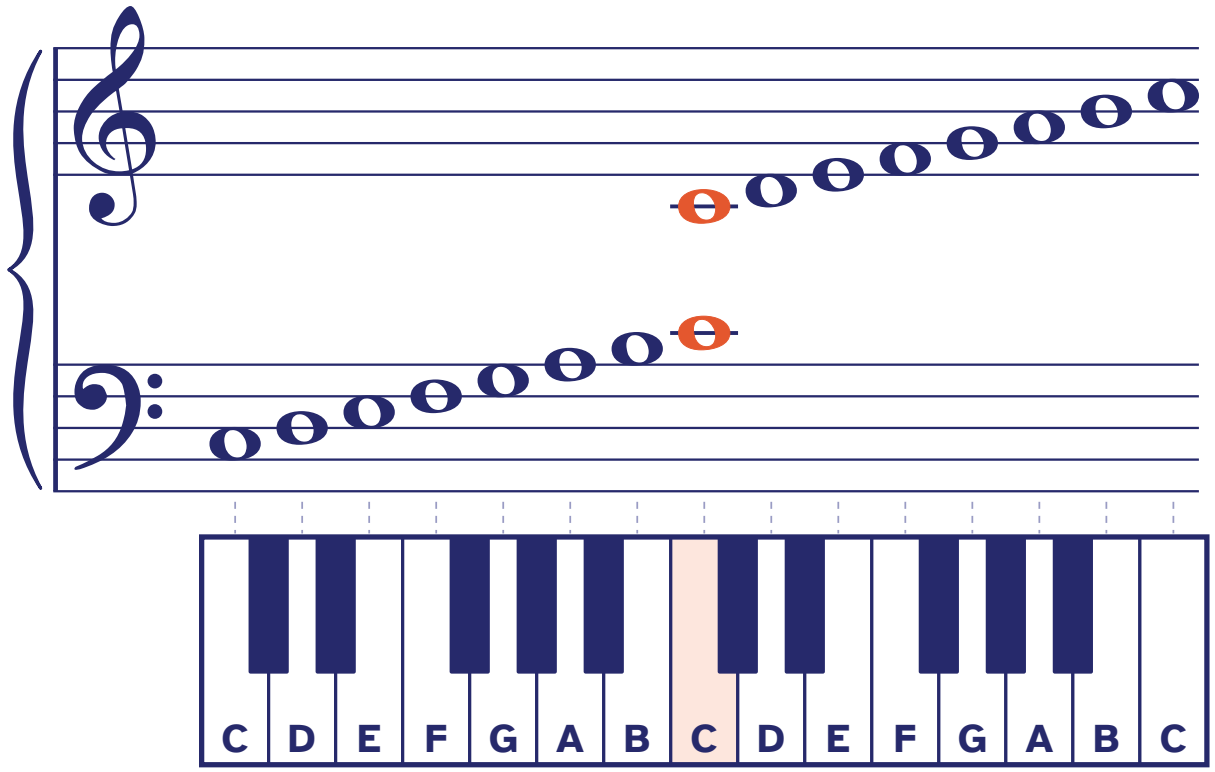
1. **The clef:** treble.
2. **A landmark:** the G the clef curls round, on line 2.
3. **Skip up** line to line, one letter jumped each time: G, B, D.

The note on line 4 is **D**.

PART I · PITCH · 4

The staff and the keyboard

Every line and space of the grand staff is one white key. Read from left to right, the notes climb the staff exactly as the keys climb the keyboard: two octaves from C3 to C5, with middle C in coral where the two staves meet.



middle C

What the map shows

Each step up the staff (line to space, space to line) is the next **white** key. The black keys have no line or space of their own: they are written with a sharp or a flat on a white key's line or space (p. 26).

SEE ALSO

The keyboard p. 22 · Reading the notes p. 24

Two ways to write one key

Middle C can sit on a ledger line below the treble staff or above the bass staff. Both mean the same key; the music chooses the staff of the hand that plays it.

PRACTISE IT

Workbook Pitch 1–2 · Online: Library › Staff and clefs
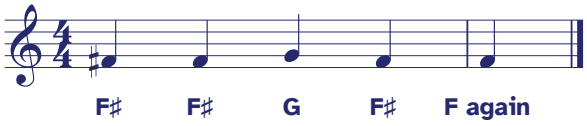
PART I · PITCH · 5

Sharps, flats and naturals

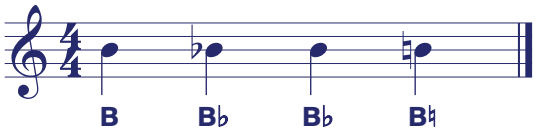
An **accidental** written before a note changes its pitch by a semitone. On the keyboard, a sharp usually means the black key to the right, a flat the black key to the left.

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	raises a note by a semitone	Level 0
♭	flat	lowers a note by a semitone	Level 0
♮	natural	cancels a sharp or flat	Level ½
𝄪	double sharp	raises by two semitones	Level 4
𝄫	double flat	lowers by two semitones	Level 4

An accidental lasts for the bar



The sharp still counts later in the bar; after the bar line, F is F again.



A natural cancels the flat for the rest of the bar.

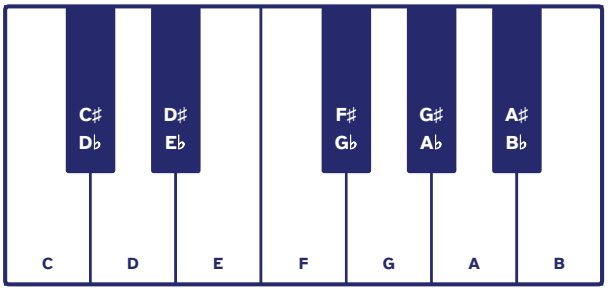
Only that line or space. A sharp on the F in the first space does not change the F an octave higher.

A tie carries it on. A note tied across the bar line keeps its accidental.

Reminders. A courtesy accidental, sometimes in brackets, just reminds you after a bar line.

One key, two names

Every black key has two names: the key between F and G is both **F♯** and **G♭**. Notes that sound the same but are written differently are **enharmonic**. White keys can have sharp or flat names too: **E♯** is F, **C♭** is B.



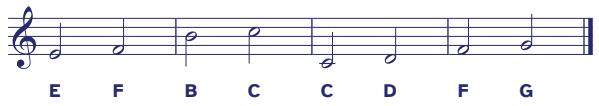
SEE ALSO
Keys and key signatures p. 31

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

PART I · PITCH · 6

Counting intervals

An **interval** is the distance between two notes. A **semitone** is the very next key; a **tone** is two semitones. An interval's **number** comes from counting letter names, both ends included.

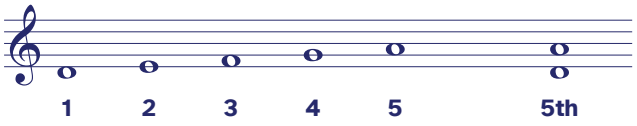


E–F and B–C are semitones; C–D and F–G are tones.



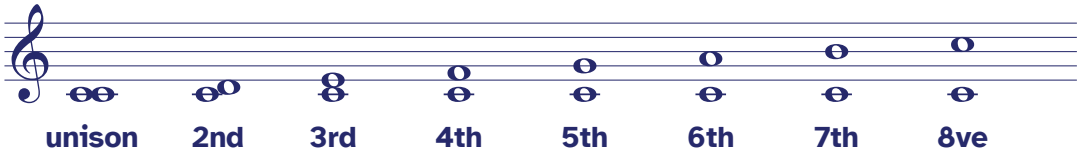
Melodic: one note after the other. Harmonic: together.

Counting the number



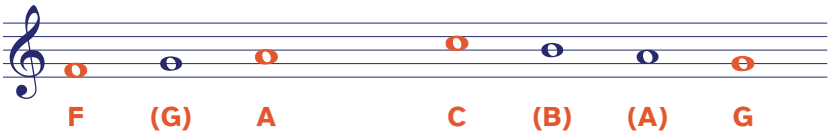
D up to A: D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

Interval numbers from C



IN ACTION

How far apart are these notes?



1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.

F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

THINK OF IT Put your thumb on C and your little finger on G: five fingers on five white keys, a 5th. Your hand measures intervals for you.

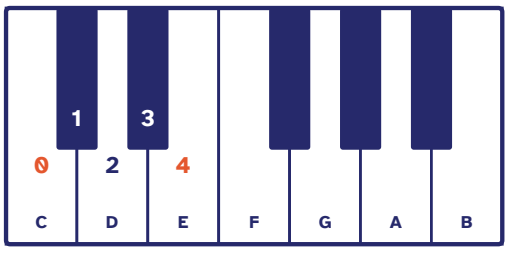
SEE ALSO
Major, minor and perfect p. 28 · Scales p. 29

PRACTISE IT
Workbook Pitch 4 · Online: Library › Intervals

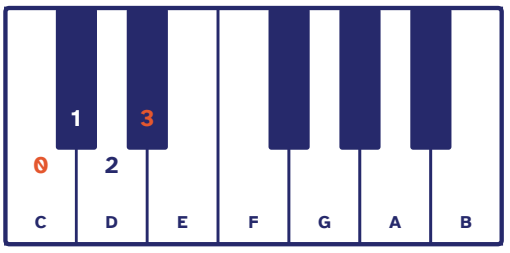
PART I · PITCH · 7

Major, minor and perfect

Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.



C to E: count every key, black and white. 4 semitones: a major 3rd.



C to E♭: 3 semitones: a minor 3rd.



Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison	0			Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** › augmented

unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** › augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.
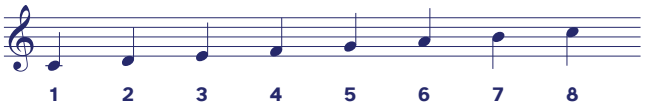
SEE ALSO
Counting intervals p. 27 · Chords p. 50

PRACTISE IT
Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · 8

Major scales

A **scale** is a ladder of notes from one key note up to the next, in a fixed pattern of tones (T) and semitones (S). The pattern, not the starting note, makes a scale major or minor.



C major: T T S T T T S. The semitones fall between 3–4 and 7–8.

SCALE	STEPS	FIRST MET
major	T T S T T T S	Level ½
natural minor	T S T T S T T	Level ½
harmonic minor	T S T T S T+S S	Level ½
melodic minor (up)	T S T T T T S; down as the natural minor	Level 2
chromatic	every semitone	Level 2
pentatonic (black keys)	T T T+S T T+S	Level 0

Degree names

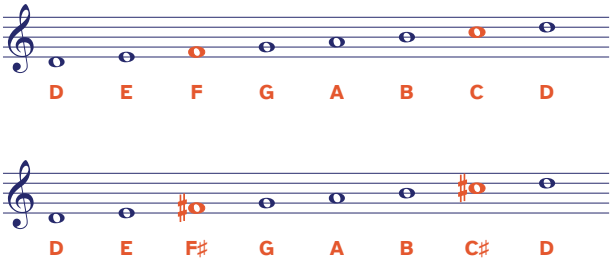
1 tonic · **2** supertonic · **3** mediant · **4** subdominant · **5** dominant · **6** submediant · **7** leading note

Major or minor?

The same key note can start a major or a minor scale: C major and C minor. The pattern decides. Major sounds bright and settled; minor, darker.

IN ACTION

Build D major



1. **Write the eight letters** from D to D, every letter once.
2. **Check the steps** against T T S T T T S: E–F is too small, B–C is too small.
3. **Raise F and C** a semitone to fix them.
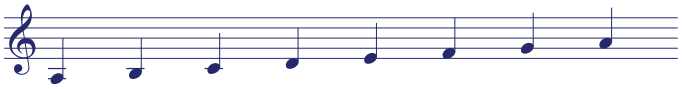
D major: **D E F♯ G A B C♯ D**, two sharps.

PART I · PITCH · 9
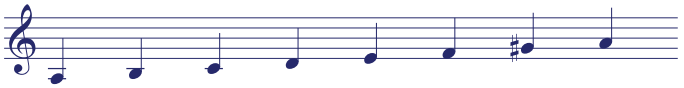
Minor scales

Every minor scale starts from the natural minor, the notes of its relative major from the 6th degree up. Two small changes give the harmonic and melodic forms, which sound stronger at the top.
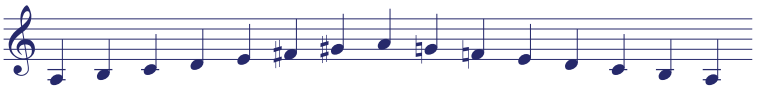
A minor in three forms



Natural: the notes of C major, from A to A. T S T T S T T.



Harmonic: the 7th raised a semitone (G♯). The gap F–G♯ is a tone and a half.



Melodic: the 6th and 7th raised going up (F♯ G♯), lowered coming down.

Relative minor

Same key signature, three semitones below the major: C major and A minor share every note.

Parallel minor

Same key note, different signature: C major (no sharps or flats) and C minor (three flats).

THINK OF IT The relative minor lives in the same house as its major, with the same furniture (notes); the parallel minor lives at the same address (key note) but furnishes it differently.

SEE ALSO
Scales p. 29 · Keys and key signatures p. 31

PRACTISE IT
Workbook Pitch 5 · Online: Scale finder, Library › Scales

PART I · PITCH · 10

Keys and key signatures

A **key signature** at the start of each line lists the sharps or flats the piece uses throughout, so they do not have to be written before every note. They always appear in the same order.



Sharps: F C G D A E B. The key note is a semitone above the last sharp.

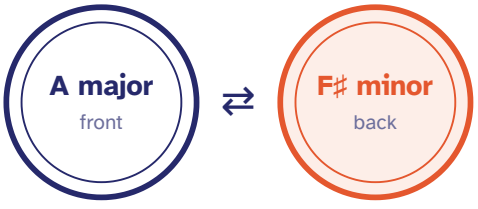


Flats: B E A D G C F. From two flats, the second-last flat names the key.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

MAJOR	RELATIVE MINOR	SIGNATURE	FIRST MET
C major	A minor	—	Level ½
G major	E minor	1♯	Level ½
D major	B minor	2♯	Level 1
A major	F♯ minor	3♯	Level 3
E major	C♯ minor	4♯	Level 4
B major	G♯ minor	5♯	Level 4
F major	D minor	1♭	Level ½
B♭ major	G minor	2♭	Level 1
E♭ major	C minor	3♭	Level 3
A♭ major	F minor	4♭	Level 4
D♭ major	B♭ minor	5♭	Level 4

Six and seven sharps or flats (F♯, C♯, G♭, C♭ major and their minors) come at Levels 4–6.



One signature, two keys

Every key signature belongs to **one major key and one minor key**, like the two sides of a coin: three sharps are A major or F♯ minor. The minor is the **relative minor**, three semitones below the major.

SEE ALSO
Scales p. 29 · Finding the key p. 32 · The circle of fifths p. 33

PRACTISE IT
Workbook Pitch 5 · Online: Scale finder, Library › Keys

PART I · PITCH · 11

Finding the key

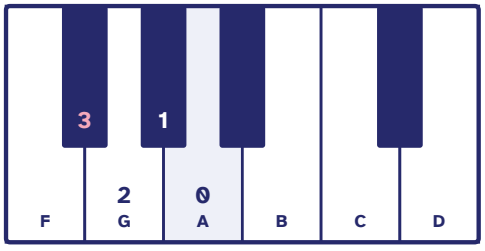
A key signature names two keys, a major and its relative minor. Three quick rules find the major; one more finds the minor; then the music itself tells you which of the two it is.



Sharps: the last sharp, then one semitone up.
Last sharp G♯ → up a semitone → **A major**.



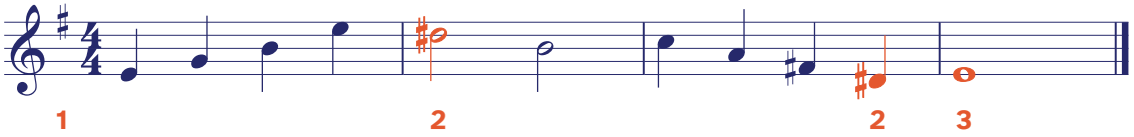
Flats: the second-last flat is the key.
B♭ E♭ A♭ → **E♭ major**. One flat alone is F major.



Major to minor: three semitones down.
A, G♯, G, F♯ → **F♯ minor**, the relative minor of A major (count every key, black and white).

IN ACTION

What key is this tune in?



1. **The key signature:** one sharp, so G major or its relative, E minor.
2. **The clue notes:** D♯ keeps appearing. D♯ is the raised 7th of E minor; G major has no D♯.
3. **The last note:** E, the key note of E minor.

The key: **E minor**. Check it the same way with any piece you play.

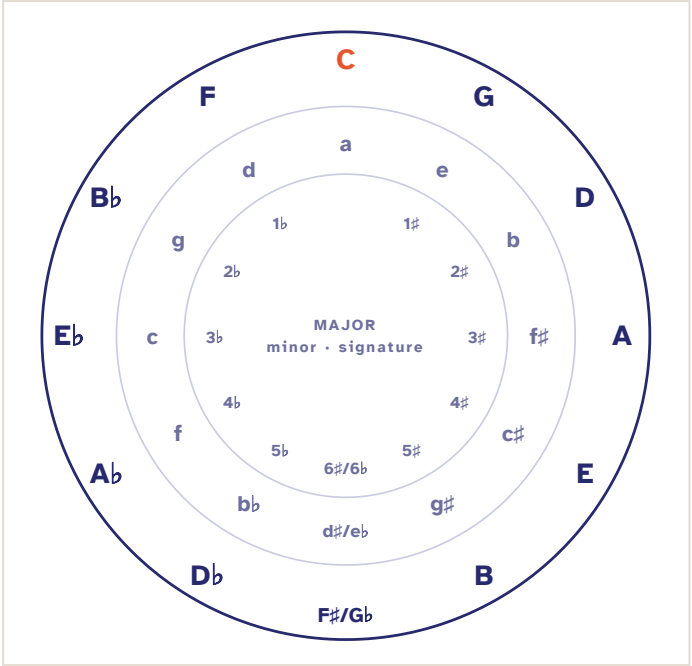
SEE ALSO
Keys and key signatures p. 31 · The circle of fifths p. 33

PRACTISE IT
Workbook Pitch 5 · Online: Scale finder, Library › Keys

PART I · PITCH · 12

The circle of fifths

All the keys on one clock face. Going round clockwise, each key is a **5th higher** than the last and has **one more sharp** (or one fewer flat). Anticlockwise, each is a 4th higher, a 5th lower, with one more flat.



Reading the circle

Outer ring: major keys. **Middle ring:** their relative minors (small letters). **Inner ring:** the key signature.

Neighbours are close

Keys next to each other share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor. These are the **related keys**.

At the bottom

Six sharps and six flats meet: F♯ major and G♭ major are the same keys on the piano, written two ways.

IN ACTION

Why do G major and D major sound so close?



G major: one sharp



D major: two sharps

1. **Find both on the circle:** D is one step clockwise from G.
2. **Compare the notes:** G A B C D E F♯ and D E F♯ G A B C♯.
3. **Only one note differs:** C becomes C♯.

Neighbours on the circle share six of their seven notes, which is why music moves between them so easily.

SEE ALSO

Keys and key signatures p. 31 · Finding the key p. 32 · Chords p. 50

PRACTISE IT

Workbook Pitch 5 · Online: Scale finder, Library › Related keys

PART II

Time

Music moves through time. Underneath it runs a steady **pulse**; the pulse is grouped into **bars**; and every note and rest takes a measured share of it. This part goes from that big picture down to the length of a single note.

1		2		3		4		5
Pulse the steady beat underneath	›	**Bar** beats in groups of 2, 3 or 4	›	**Time signature** how many beats, and which note is one beat	›	**Note values** how long each sound lasts	›	**Rhythm** the pattern of long and short

IN THIS PART

Pulse, tempo and the bar 36
Notes and rests 37
How long is a note? 38
Time signatures 39
Compound and irregular time 40
Rhythm: groups and upbeats 41
Rhythm: off the beat 42

PART II · TIME · 1

Pulse, tempo and the bar

Under almost every piece runs a **pulse**: a steady beat, like a heartbeat or a clock. Its speed is the **tempo**. Beats are grouped into **bars**, and the first beat of every bar is the strongest.

Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Tempo words such as *Andante* and *Allegro* are on page 70.

Strong and weak beats

BEATS	PATTERN	FEELS LIKE
2	**Strong** weak	a march
3	**Strong** weak weak	a waltz
4	**Strong** weak **medium** weak	most songs

Bars and bar lines

SIGN	NAME	MEANING
	bar line	divides the music into bars
	double bar line	marks the end of a section, or a change of key, time signature or tempo
	final bar line	marks the end of the piece
	repeat signs	play the passage between them again; if there is no start sign, go back to the beginning

Counting



Three bars of 3/4. Every beat is counted, even inside a long note: the coral number is where you play, the grey numbers are the beats you hold.

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

WATCH OUT

- × Speeding up in the easy bars and slowing down in the hard ones: choose a tempo you can keep in the hardest bar.
- × Counting the notes instead of the beats.

SEE ALSO
Time signatures p. 39 · Tempo words p. 70

PRACTISE IT
Workbook Level 0 · Pulse · Online: Metronome

PART II · TIME · 2

Notes and rests

A note's shape shows how long it lasts: each plain note lasts **half as long** as the one above it. Every note has a **rest** of the same length, a silence counted just as carefully as a sound.

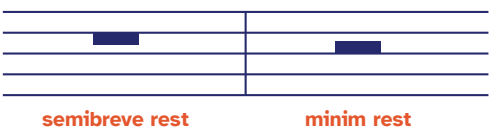
NOTE	REST	NAME	ALSO CALLED	BEATS*	FIRST MET
		breve	double whole note	**8**	Level 4
		dotted semibreve	dotted whole note	**6**	*Extra*
		semibreve	whole note	**4**	Level 0
		dotted minim	dotted half note	**3**	Level 0
		minim	half note	**2**	Level 0
		dotted crotchet	dotted quarter note	**1½**	Level ½
		crotchet	quarter note	**1**	Level 0
		dotted quaver	dotted eighth note	**¾**	Level 3
		quaver	eighth note	**½**	Level 0
		semiquaver	sixteenth note	**¼**	Level 2
		demisemiquaver	32nd note	**⅛**	*Extra*
		hemidemisemiquaver	64th note	**1/16**	*Extra*

* With the crotchet as one beat, as in 2/4, 3/4 and 4/4. *Extra*: beyond Level 8.

Parts of a note

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

Which rest is which?
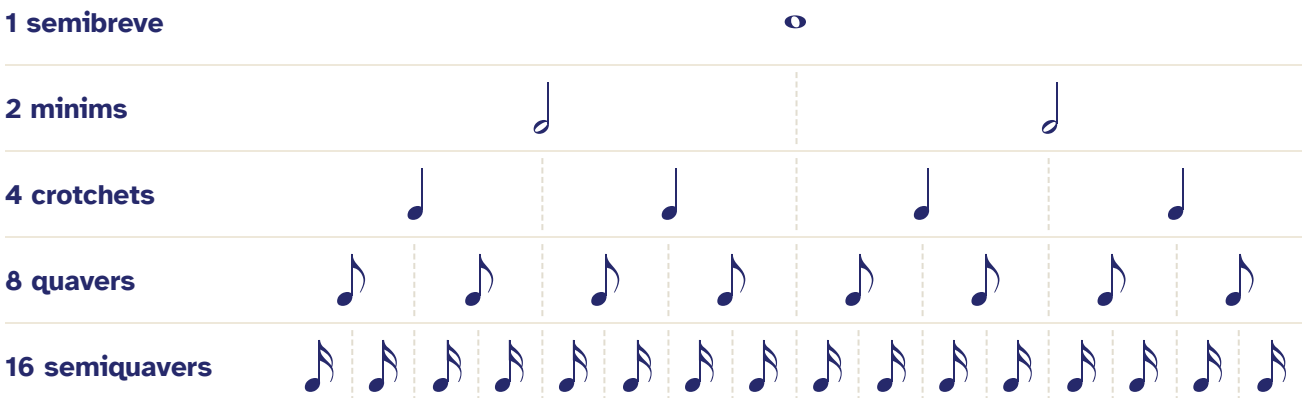


The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

PART II · TIME · 3
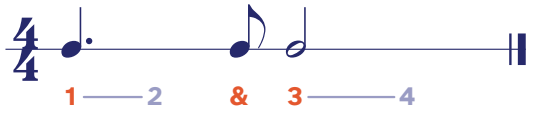
How long is a note?

Each row lasts exactly as long as the row above it: the same stretch of time, cut in half again and again.



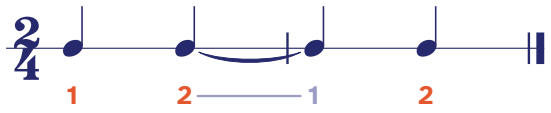
The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½. A **double dot** adds half, then a quarter: a double-dotted minim lasts 2 + 1 + ½ = 3½.



The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.



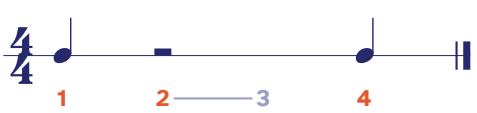
TIE OR SLUR?

The same curve over notes of **different** pitches is a **slur**: play them smoothly joined (legato). Same pitch: a tie. Different pitches: a slur.

Rests in a bar



✓ The silence on beats 3–4 is one minim rest; a whole silent bar takes a semibreve rest, in any time signature.



× In 4/4, a minim rest on beats 2–3 hides the middle of the bar: write two crotchet rests.

SEE ALSO

Time signatures p. 39 · Rhythm p. 41

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests

PART II · TIME · 4

Time signatures

Two numbers at the start of the music. The **top number** says how many counts are in each bar; the **bottom number** says which note gets one count.

The bottom number

BOTTOM	NOTE	ONE COUNT IS A	FIRST MET
2		minim	Level 3
4		crotchet	Level 0
8		quaver	Level 1
16		semiquaver	*Extra*

SIMPLE OR COMPOUND?

In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

Quick test: a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

Simple time: each beat divides in two

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
2/4	2 crotchets	Level 0
3/4	3 crotchets	Level 0
4/4 	4 crotchets (the C sign: common time)	Level 0
2/2 	2 minims (the cut C: cut common time)	Level 3
3/2	3 minims	Level 3
4/2	4 minims	*Extra*
3/8	3 quavers	Level 3

Changing time signature

A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

WATCH OUT

- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (page 40).

SEE ALSO

Pulse and the bar p. 36 · Compound time p. 40

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

PART II · TIME · 5
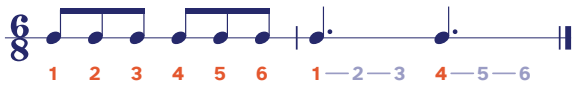
Compound and irregular time

Compound time has dotted beats that divide into three; irregular time has five or seven counts, grouped unevenly. (9/4, 12/4 and the /16 signatures exist but are rare.)

Compound time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
6/8	2 dotted crotchets	Level 1
9/8	3 dotted crotchets	Level 3
12/8	4 dotted crotchets	Level 3
6/4	2 dotted minims	Level 4

Counting 6/8



Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

3/4 or 6/8?



3/4: six quavers in three twos



6/8: six quavers in two threes

Irregular time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
5/4	5 crotchets: 3 + 2 or 2 + 3	Level 0
7/4	7 crotchets: 4 + 3 or 3 + 4	*Extra*
5/8	5 quavers: 3 + 2 or 2 + 3	*Extra*
7/8	7 quavers: 2 + 2 + 3 or 3 + 2 + 2	*Extra*

UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: “1 2 3, 1 2 3, 1 2”.
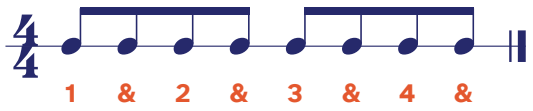


PART II · TIME · 6

Rhythm: groups and upbeats

Rhythm is the pattern of long and short sounds over the pulse. The pulse stays steady; the rhythm moves around it. The way notes are grouped on the page shows where the beats are.

Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar. In compound time, quavers are beamed in threes.



✓ The middle of the bar can be seen.



× A beam across beats 2–3 hides it.

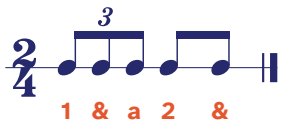
The upbeat (anacrusis)

A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.
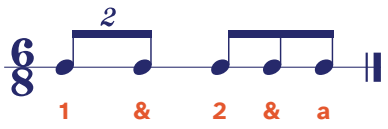


Count “1 2” silently, then play on 3.

Triplets and duplets



Triplet: three notes in the time of two.

Level 2. Three exactly equal notes.



Duplet: two notes in the time of three, in compound time.

Level 2.

Counting smaller values



Crotchet, two quavers, four semiquavers and a triplet (“1 & a”): one beat each.

PART II · TIME · 7

Rhythm: off the beat

Once the pulse is secure, rhythm can pull against it. Every one of these effects needs a steady beat underneath.

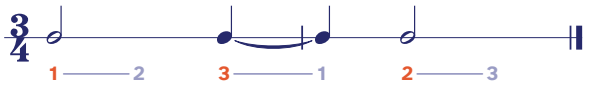
Syncopation



The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

Hemiola



Two bars of 3/4 felt as three minims.

Level 4. Common at cadences in Baroque dances.

Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.

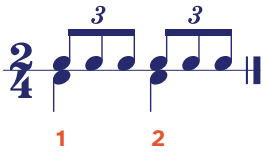


Written



Played

Polyrhythm Level 7



Three against two: a triplet above, two crotchets below, starting together.

Rubato Extra

Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much Romantic music.

WATCH OUT

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

SEE ALSO
Notes and rests p. 37 · Words for how to play p. 70

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock